

Endocrine pathology

Dr. Hidayat Y. (M.D)

Learning Objectives

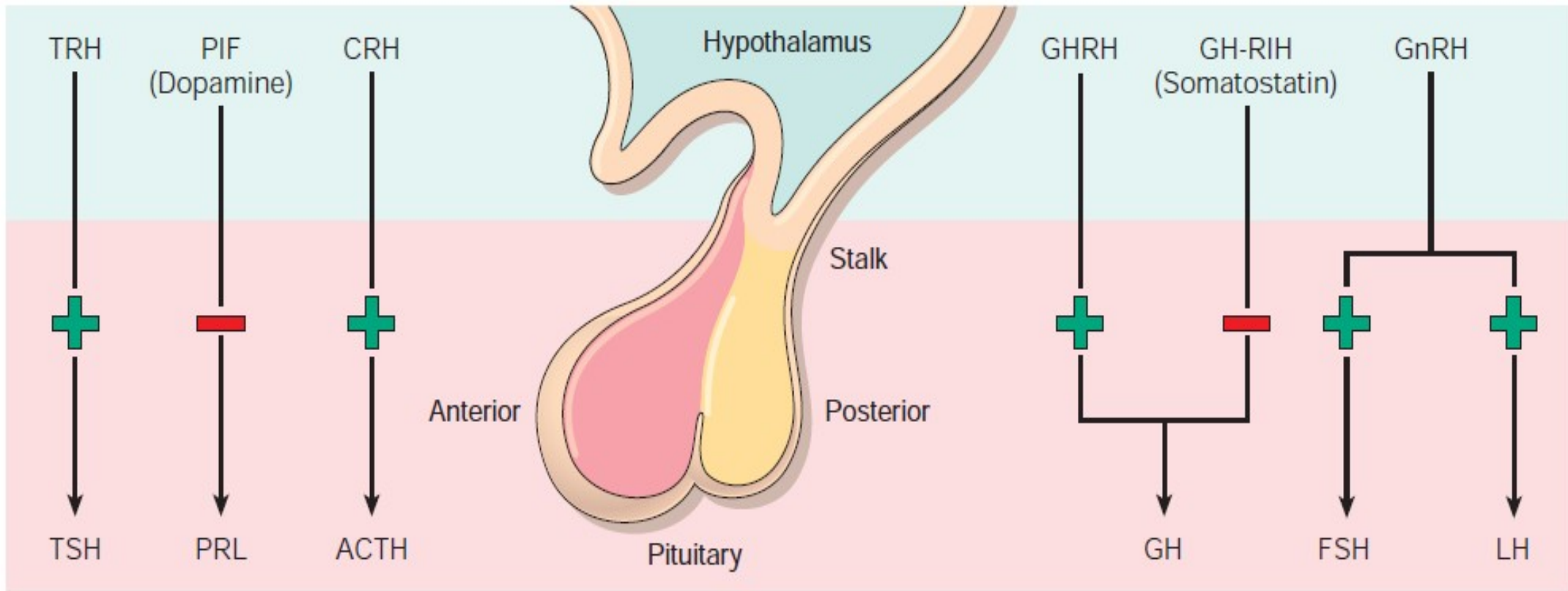
- Demonstrate understanding of disease of the thyroid, parathyroid, adrenal and pituitary
- Describe the relationship between the hypothalamus and pituitary gland
- Solve problems concerning multiple endocrine neoplasia syndromes
- Explain information related to diabetes mellitus

Introduction

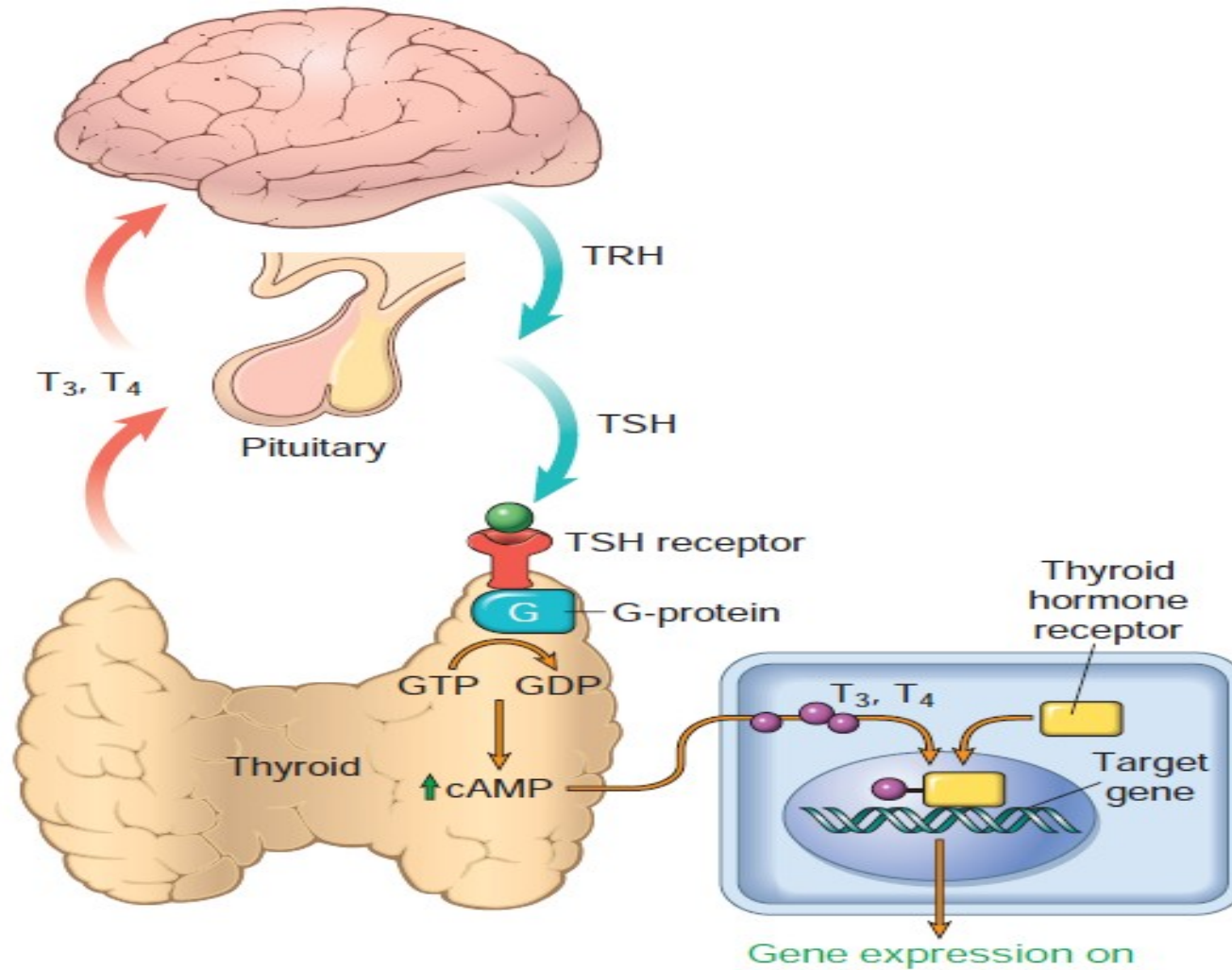
- The endocrine system consists of a highly integrated and widely distributed group of organs, called glands,
- Orchestrate a state of metabolic equilibrium among the various organs of the body.
- Three types of signaling by secreted molecules
 - autocrine,
 - paracrine,
 - Endocrine.

Pituitary Gland

- small, bean-shaped structure that lies at the base of the brain within the sella turcica.
- Its function is controlled by the hypothalamus
- The production of several hormones from endocrine glands is stimulated by **trophic factors** released from the pituitary.
- The endocrine hormone inhibits production of the trophic factors, a process known as *feedback inhibition*, thus maintaining physiologic levels of the hormone.
- **The anterior pituitary, or adenohypophysis, which constitutes about 80% of the gland, produces trophic hormones that stimulate the production of hormones from the thyroid, adrenal, and other glands.**
 - *Somatotrophs* produce growth hormone (GH).
 - *Lactotrophs* produce PRL.
 - *Corticotrophs* produce adrenocorticotrophic hormone (ACTH), pro-opiomelanocortin (POMC), and melanocyte stimulating hormone (MSH).
 - *Thyrotrophs* produce thyroid-stimulating hormone (TSH).
 - *Gonadotrophs*-FSH and LH



Thyroid gland



The net result is an increase in the basal metabolic rate

Hyperthyroidism

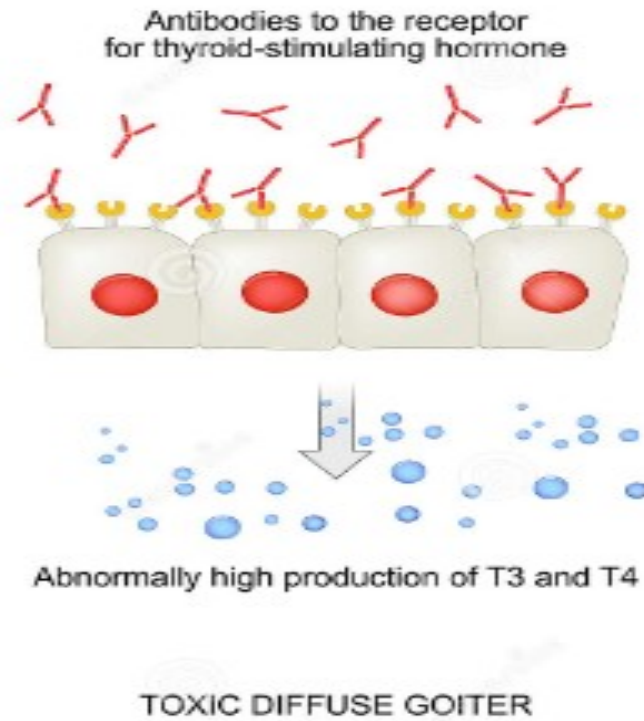
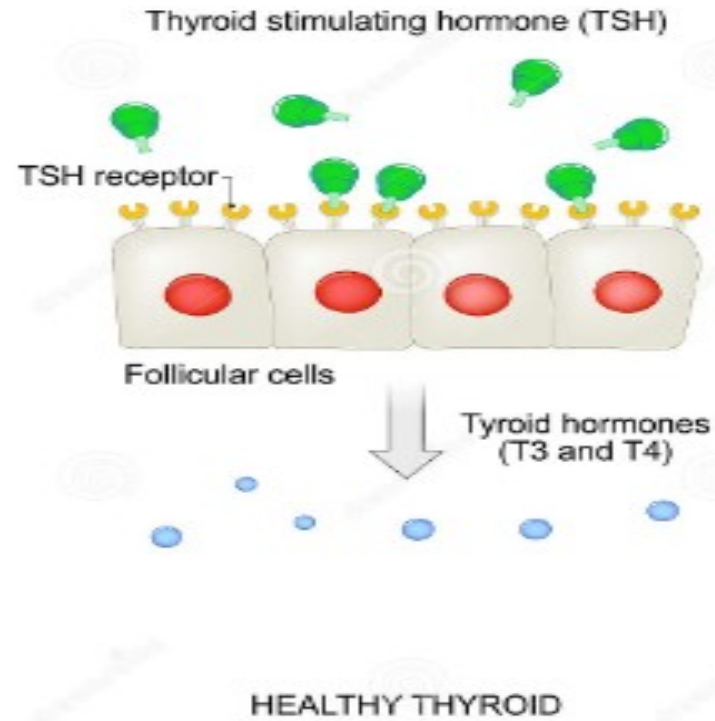
- The term hyperthyroidism is used when the mean metabolic rate of all cells is increased due to increased T4 or T3.
- **Thyrotoxicosis** is a hypermetabolic state caused by elevated circulating levels of free T3 and T4.
 - Could be primary hyperthyroidism or secondary hyperthyroidism.
- **The most common causes of thyrotoxicosis** are associated with hyperfunction of the gland and include the following:
 - *Diffuse hyperplasia* of the thyroid associated with Graves disease (approximately 85% of cases)
 - Hyperfunctional *multinodular goiter*
 - Hyperfunctional thyroid *adenoma*

- **Clinical features** include tachycardia and palpitations; nervousness and diaphoresis; heat intolerance; weakness and tremors; diarrhea; and weight loss despite a good appetite.
- Lab studies:
 - In primary hyperthyroidism, TSH is decreased.
 - In secondary and tertiary hyperthyroidism, TSH is elevated.
 - Free T4 is elevated
- ✓ NB. **TSH** is the most sensitive test in thyroid disease. If it is normal, the patient is euthyroid.

Graves disease

- is an autoimmune disease characterized by production of IgG autoantibodies to the TSH receptor.
- Females >males
- peak age 20–40.
- The disease is characterized by a triad of clinical findings:
 - *Hyperthyroidism* associated with diffuse enlargement of the gland
 - Infiltrative *ophthalmopathy* and resultant exophthalmos
 - Localized, infiltrative *dermopathy*, sometimes called *pretibial myxedema*, (minority of patients)
- Laboratory- Increase in T3,T4 and decrease in TSH
- Microscopically, the thyroid has hyperplastic follicles with scalloped colloid.

Graves' disease (autoimmune disease)



DIFFUSE AND MULTINODULAR GOITER

- Enlargement of the thyroid, or goiter, is caused by impaired synthesis of thyroid hormone, which is most often the result of **dietary iodine deficiency**.
- Reduced thyroid hormone → compensatory rise in the serum TSH level → hypertrophy and hyperplasia of thyroid follicular cells → enlargement of the thyroid gland.
- Females are affected more often than males.
- It is frequently **asymptomatic**
- Typically euthyroid, with normal T4, T3, and TSH.

Diffuse goiter

- Diffuse nontoxic (simple) goiter causes enlargement of the entire gland without producing nodularity
- low levels of iodine.
- most are clinically euthyroid.
- Clinical manifestations are primarily related to
- *mass effects* from the enlarged thyroid
- Serum T3 and T4 levels are normal,
- TSH is usually elevated or at the upper range



Multinodular Goiter

- With time, recurrent episodes of hyperplasia and involution combine to produce a more irregular enlargement of the thyroid, termed *multinodular goiter*.
- **mistaken for neoplasms than any other form of thyroid disease**
- Most patients are euthyroid or have subclinical hyperthyroidism (identified by reduced TSH levels),
- 10% of patients over a 10-year period develop an autonomous nodule within a long-standing goiter that produces hyperthyroidism (*toxic multinodular goiter*), also known as *Plummer syndrome*.
- The incidence of malignancy in long-standing multinodular goiters is low (<5%)



➤ Other causes of hyperthyroidism include

- Toxic multinodular goiter,
- Toxic adenoma (functioning adenoma producing thyroid hormone),
- Hashimoto and subacute thyroiditis (transient hyperthyroidism).

Hypothyroidism

- The term hypothyroidism is used when the mean metabolic rate of all cells is decreased due to decreased T4 or T3.
- **Clinical features** include
 - fatigue and lethargy;
 - sensitivity to cold temperatures;
 - decreased cardiac output;
 - myxedema (accumulation of proteoglycans and water);
 - facial and periorbital edema;
 - peripheral edema of the hands and feet; deep voice; macroglossia;
 - constipation; and
 - anovulatory cycles

- Lab studies:** – In primary hypothyroidism, TSH is elevated.
- In secondary and tertiary hypothyroidism, TSH is decreased.
 - Free T4 is decreased.

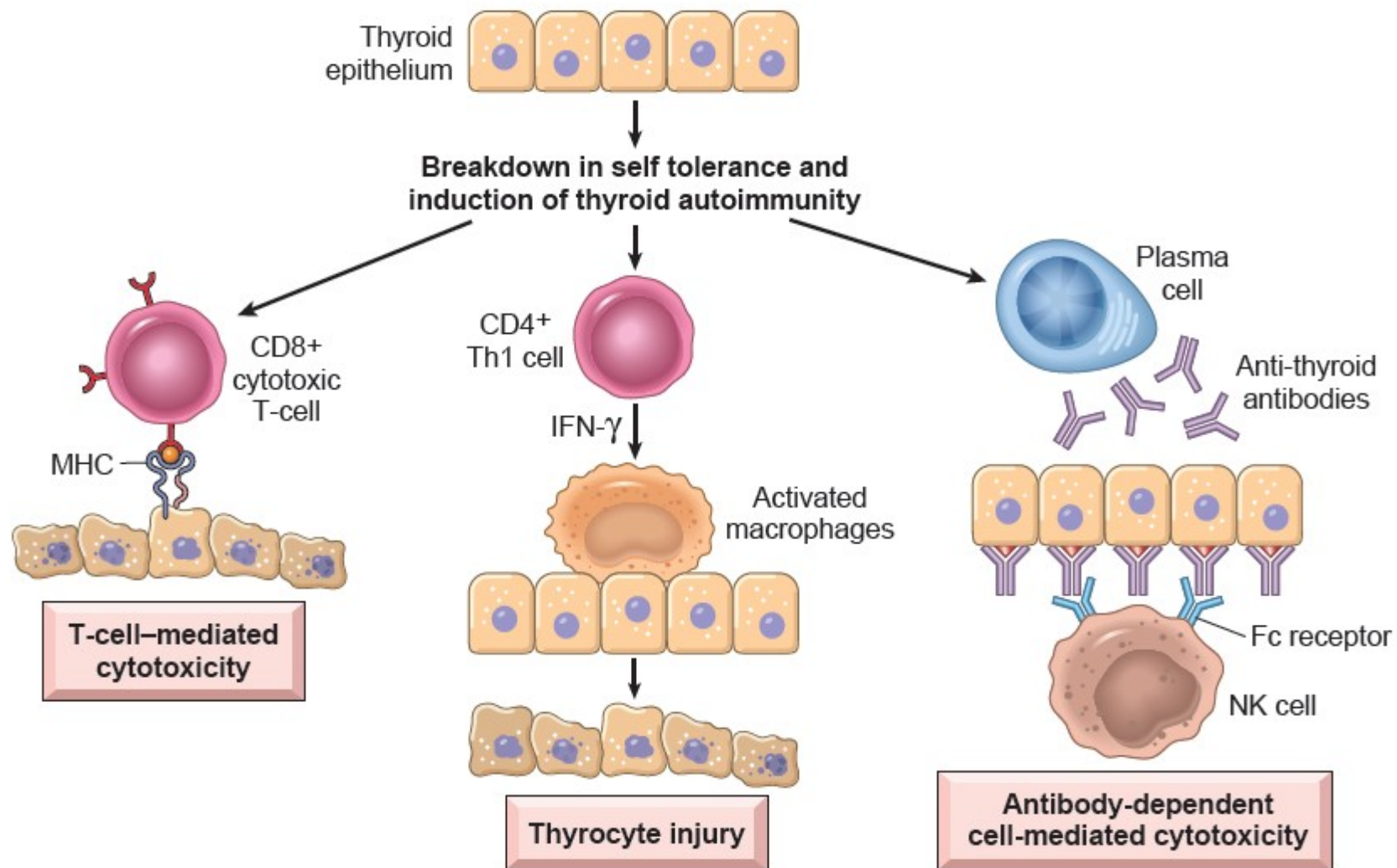
Causes of hypothyroidism

- **Iatrogenic hypothyroidism** is the most common cause of hypothyroidism in the United States, and is secondary to thyroidectomy or radioactive iodine treatment.
- **Congenital hypothyroidism** (cretinism) in endemic regions is due to **iodine deficiency** during intrauterine and neonatal life, and in non endemic regions is due to **thyroid dysgenesis**.
- Patients present with failure to thrive, stunted bone growth and dwarfism, spasticity and motor incoordination, and mental retardation.
- Goiter is seen in endemic cretinism

Thyroiditis

➤ Hashimoto thyroiditis

- is a chronic autoimmune disease characterized by immune destruction of the thyroid gland and hypothyroidism.
- It is the most common non iatrogenic/non idiopathic cause of hypothyroidism in the United States;
- it most commonly causes painless goiter in females more than males, with peak age 40–65.



- Initially may cause transient hyperthyroidism (hashitoxicosis).
- Hashimoto may be associated with other autoimmune diseases (SLE, rheumatoid arthritis, Sjögren syndrome, etc.), and it has an increased risk of non-Hodgkin B-cell lymphoma.
- Grossly, Hashimoto produces a pale, enlarged thyroid gland; microscopically, it shows lymphocytic inflammation with germinal centers and epithelial “Hürthle cell” changes.

Subacute thyroiditis (de Quervain thyroiditis/ granulomatous thyroiditis)

- Is the second most common form of thyroiditis;
- it affects females more than males, with peak age 30–50.
- Typically preceded by a viral illness
- Produces a tender, firm, enlarged thyroid gland
- May be accompanied by transient hyperthyroidism
- Microscopy shows granulomatous thyroiditis.
- The disease typically follows a self-limited course.

Thyroid Neoplasia

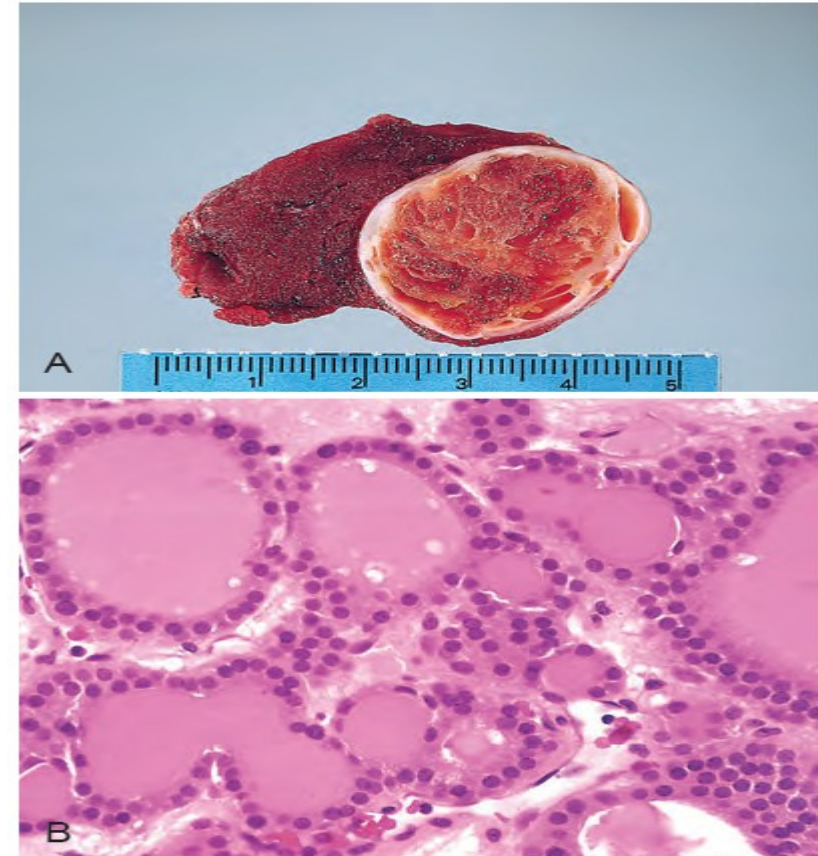
- Thyroid nodules is the possibility of a malignant neoplasm.
- Majority of solitary nodules of the thyroid → nonneoplastic lesions (e.g., a dominant nodule in multinodular
- goiter, simple cysts, or foci of thyroiditis) or benign neoplasms such as follicular adenoma.
- In fact, benign neoplasms outnumber thyroid carcinomas by a ratio of nearly 10 : 1.
- While only 1% of solitary thyroid nodules are malignant

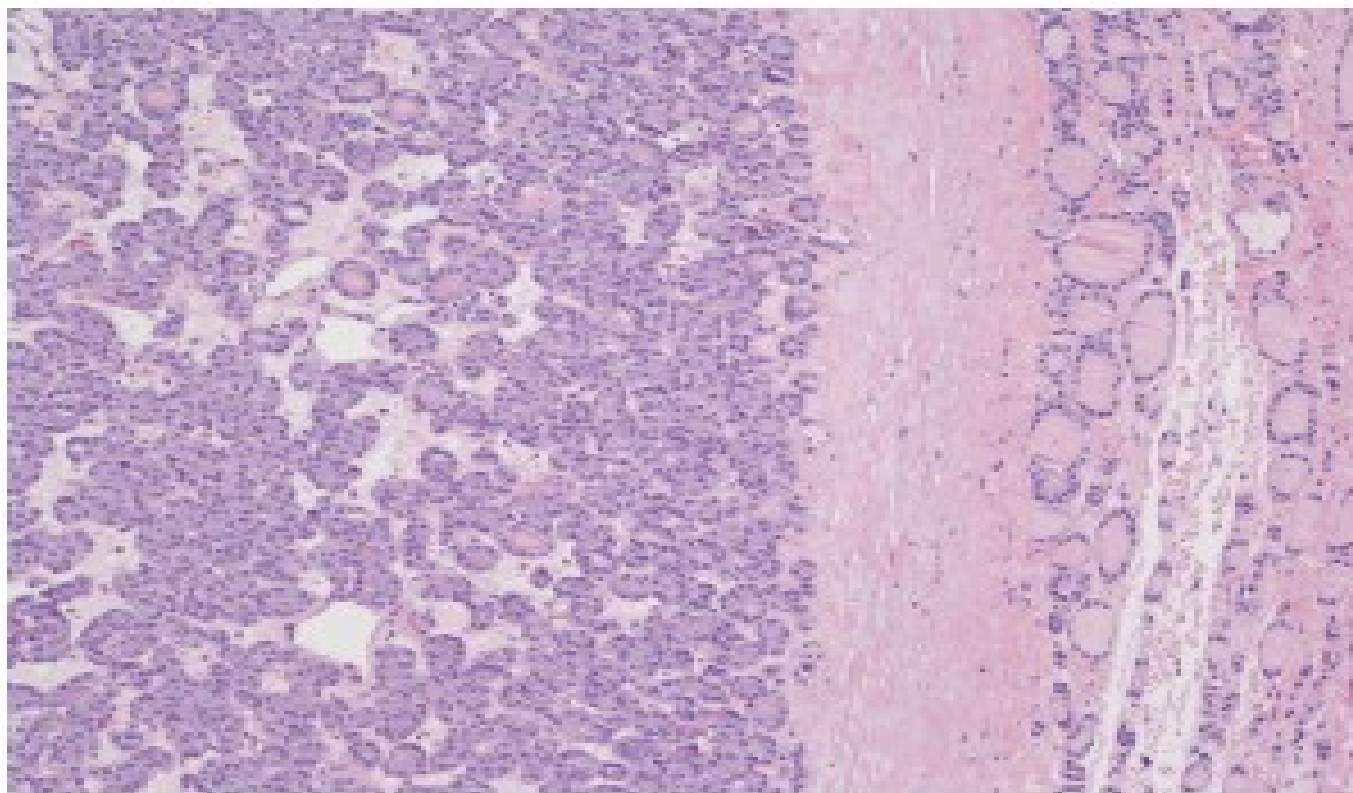
Adenomas

- Adenomas of the thyroid are typically discrete, solitary masses, derived from follicle epithelium, and hence they are also known as follicular adenomas
- vast majority of adenomas are nonfunctional
- Can not differentiate adenoma and

Carcinoma based on FNA. (Pitfall of FNA)

- Histologic examination!!!





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Figure 25-1. Follicular Adenoma (*left*), Separated from Normal Thyroid Parenchyma (*right*), by the Capsule (*center*)

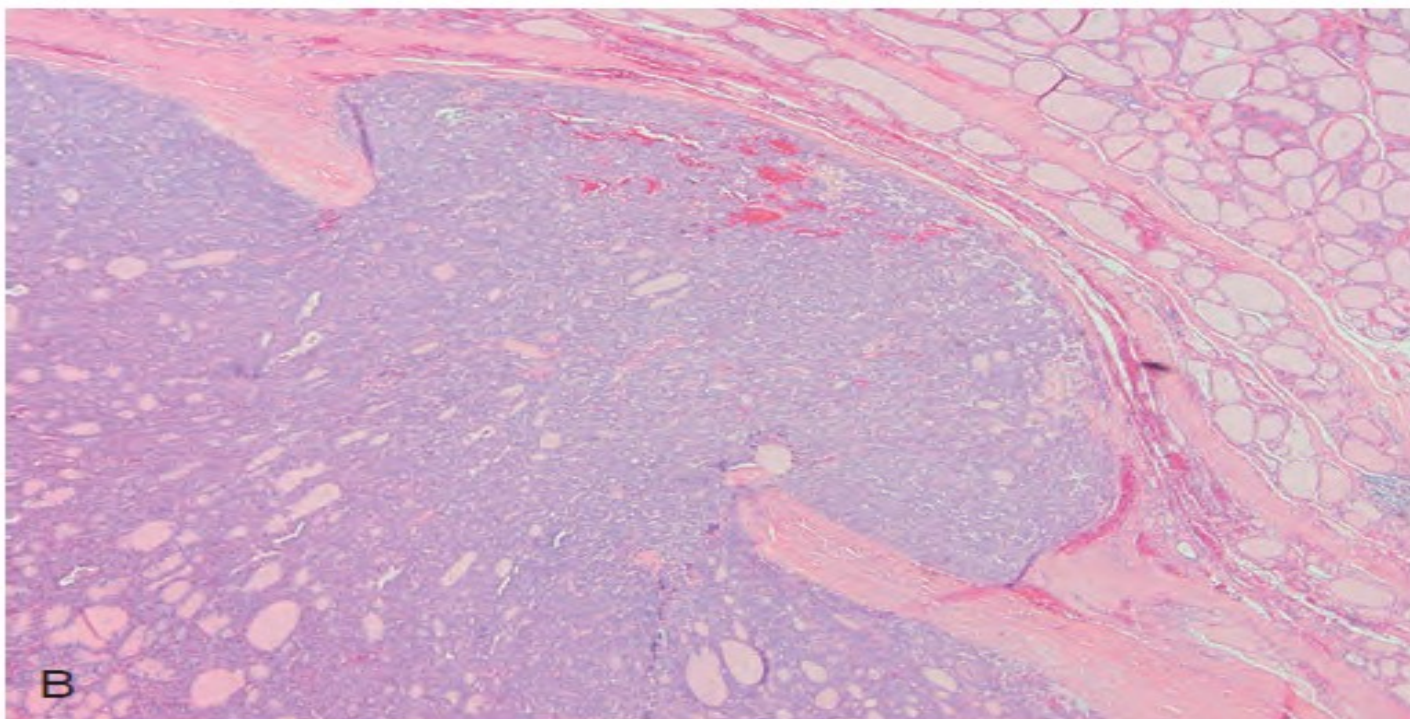
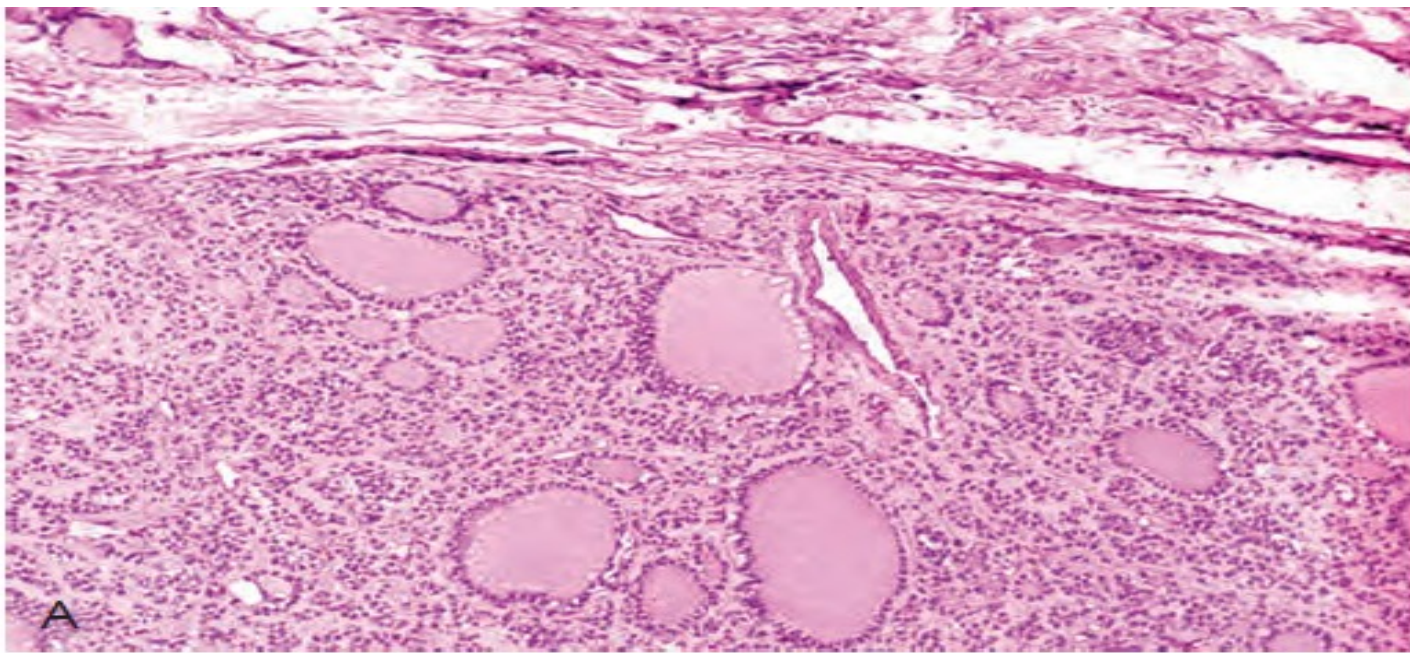
Malignant

➤ Papillary carcinoma

- accounts for 80% of malignant thyroid tumors.
- It affects females more than males, with peak age 20–50.
- Radiation exposure is a risk factor.
- The prognosis is excellent, with 20-year survival 90% due to slow growth and metastasis to regional cervical lymph nodes.
- Microscopically,
 - Papillary pattern.
 - Occasional psammoma bodies.
 - Nuclear features include clear “Orphan Annie eye” nuclei, nuclear grooves, and intranuclear cytoplasmic inclusions.
- Lymphatic spread to cervical nodes is common.

➤ Follicular carcinoma

- Accounts for 15% of malignant thyroid tumors.
- It affects females more than males, with peak age 40–60.
- Hematogenous metastasis to the bones or lungs is common.
- These cancers are microscopically distinguished from follicular adenoma by the presence of capsular invasion or vascular invasion.



➤ **Medullary carcinoma**

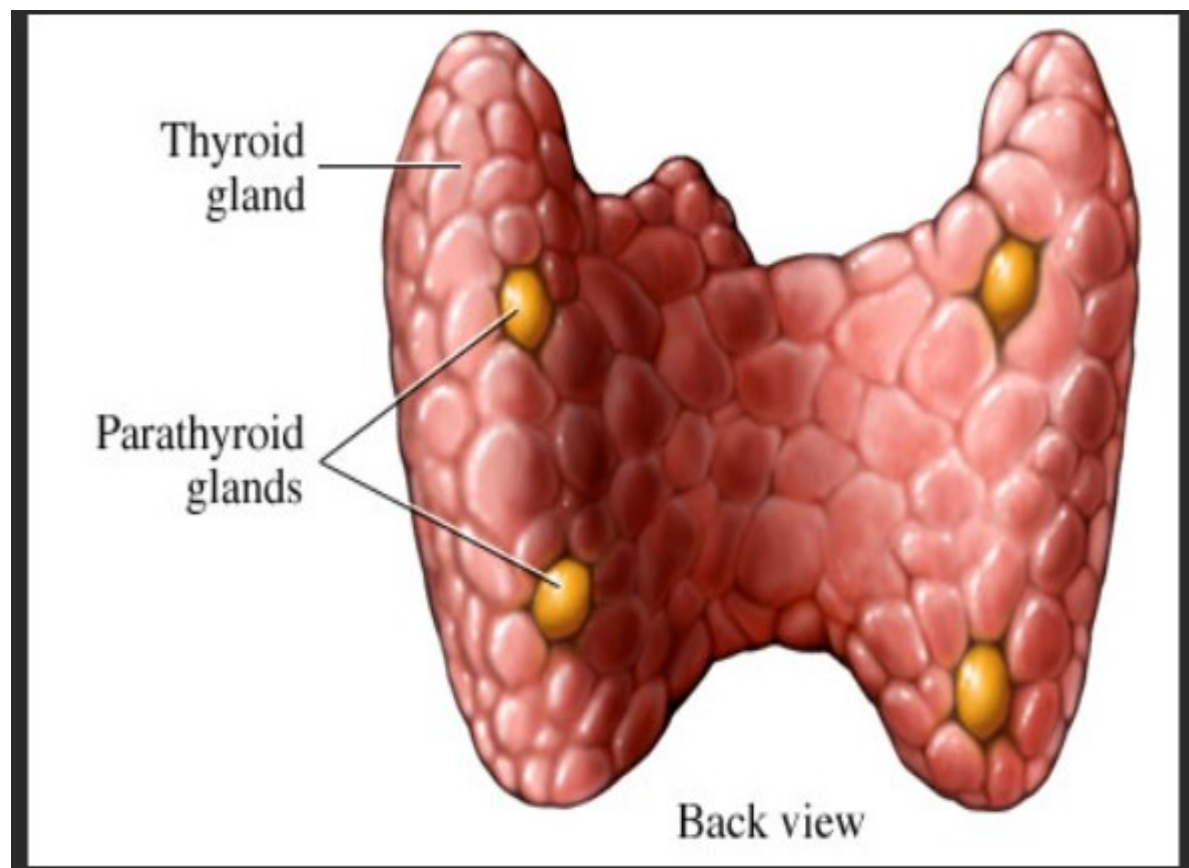
- Accounts for 5% of malignant thyroid tumors.
- It arises from C cells (**parafollicular cells**) and secretes calcitonin.
- Microscopic exam shows nests of polygonal cells in an amyloid stroma.
- A minority of cases (25%) is associated with MEN 2 and MEN 3 syndromes, and those cases tend to be multicentric.
- Activating RET mutations are present in familial and sporadic types.

➤ **Anaplastic carcinoma**

- Affects females more than males, with peak age >60.
- It can present with a firm, enlarging, and bulky mass, or with dyspnea and dysphagia.
- The tumor has a tendency for early widespread metastasis and invasion of the trachea and esophagus.
- Microscopically, the tumor is composed of undifferentiated, anaplastic, and pleomorphic cells.
- This very aggressive tumor is often rapidly fatal.

PARATHYROID GLANDS

- They are most commonly located in close proximity to the upper and lower poles of each thyroid lobe.
- composed of two cell types: chief cells and oxyphil cells
- Chief cells have secretory granules containing *parathyroid hormone (PTH)*.
- **The function of the parathyroid glands is to regulate calcium homeostasis.**
- Normally, decreased levels of free calcium stimulate the synthesis and secretion of PTH.



Hyperparathyroidism

1. Primary hyperparathyroidism

- autonomous overproduction of PTH
- can be caused by the following:
 - ✓ Adenomas (80%); may be associated with MEN 1
 - ✓ Parathyroid hyperplasia (15%) – Characterized by diffuse enlargement of all 4 glands; – Parathyroid carcinoma is very rare
 - ✓ Hyperparathyroidism can also occur as a paraneoplastic syndrome of lung and renal cell carcinomas.
- ❖ The excess production of parathyroid hormone (PTH) leads to **hypercalcemia**, with lab studies showing elevated serum calcium and PTH.

- Hyperparathyroidism is often asymptomatic, but may cause kidney stones, osteoporosis and osteitis fibrosa cystica, metastatic calcifications, or neurologic changes.

2. Secondary hyperparathyroidism

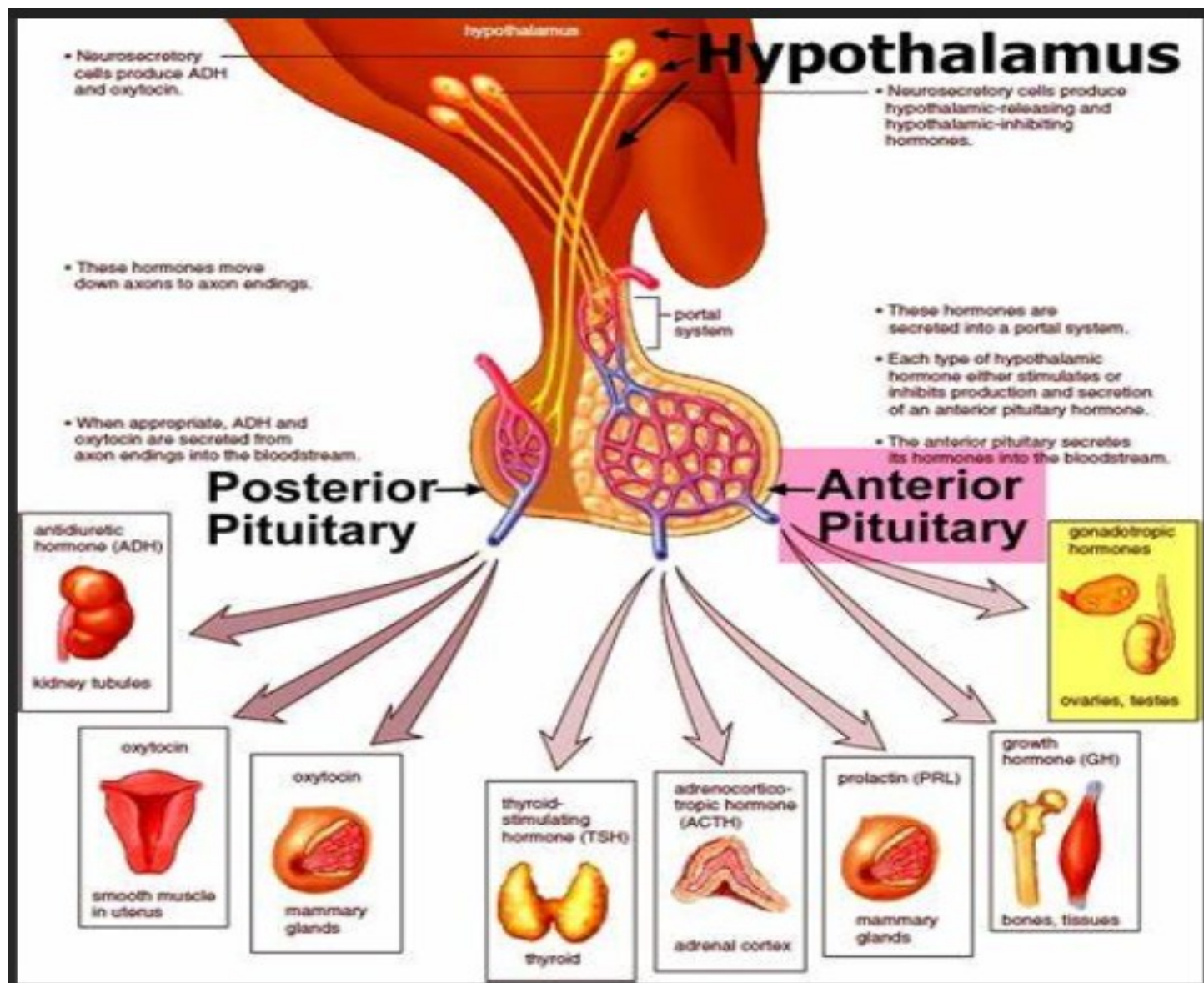
- Is caused by any disease that results in hypocalcemia, leading to increased secretion of PTH by the parathyroid glands;
- It can result from chronic renal failure-Most common
- Vitamin D deficiency, or malabsorption

Hypoparathyroidism

- Can result from the surgical removal of glands during thyroidectomy, **DiGeorge syndrome**, or a hereditary autoimmune syndrome caused by mutations in the autoimmune regulator gene AIRE.
- Patients present with muscle spasm and tingling of toes and lips.
- The hypocalcemia may also cause psychiatric disturbances and cardiac conduction defects (ECG: prolonged QT interval).

Pituitary Glands

- **Pituitary adenoma:** are categorized as follows:
 - ✓ Microadenoma if <1 cm
 - ✓ Macroadenomas if >1 cm -cause visual field defects.
 - GNAS1 mutations are common



- ✓ **Prolactinoma** is the most common type of pituitary adenoma.
 - **Lactotroph cells** secrete prolactin, which results in hyperprolactinemia.
 - Clinical features include galactorrhea, amenorrhea, and infertility, or decreased libido and impotence.
- ✓ **Nonfunctional adenoma** may produce hypopituitarism.
- ✓ **Growth-hormone–producing adenoma** is characterized by elevated growth hormone (GH) and elevated somatomedin C (insulin-like growth factor 1 [IGF-1]).
 - It causes gigantism in children and acromegaly in adults

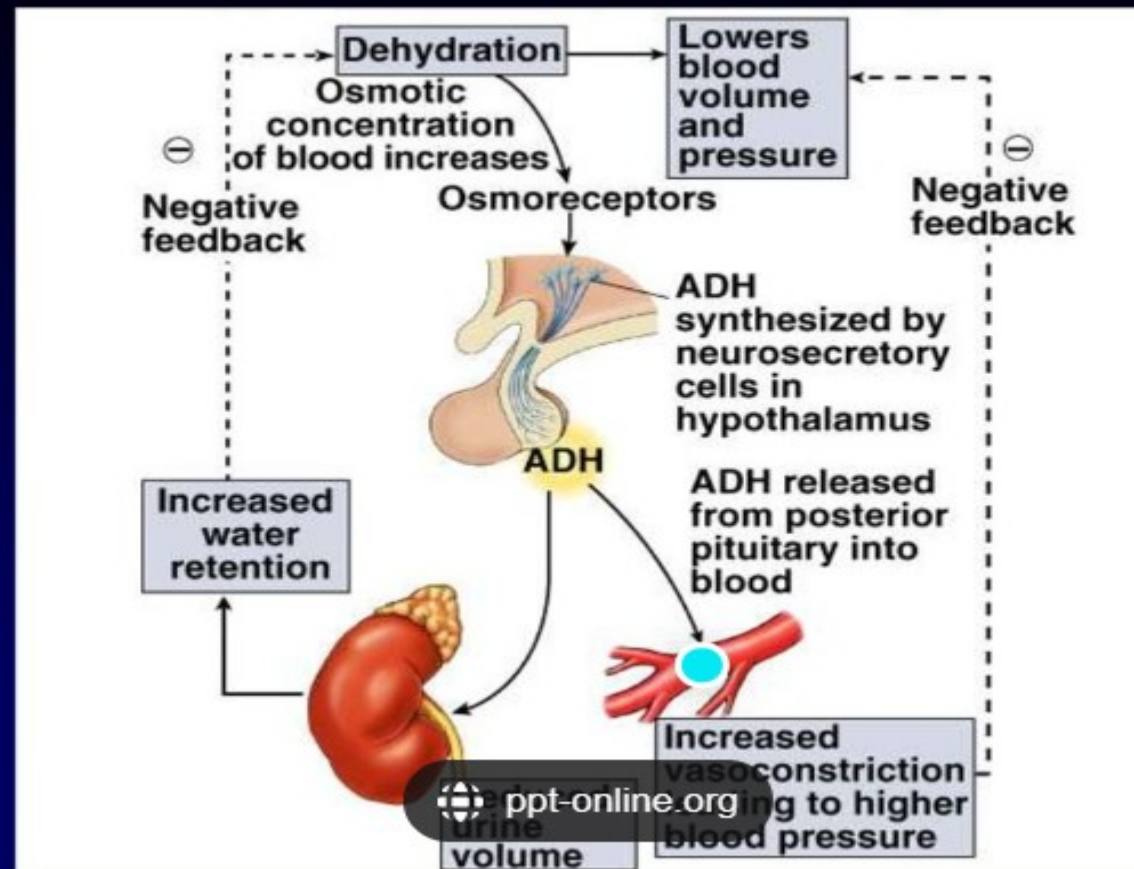
Sheehan syndrome

- Is ischemic necrosis of the pituitary secondary to hypotension from postpartum hemorrhage resulting in pan hypopituitarism.

Posterior pituitary syndromes

Diabetes insipidus

Effects of ADH



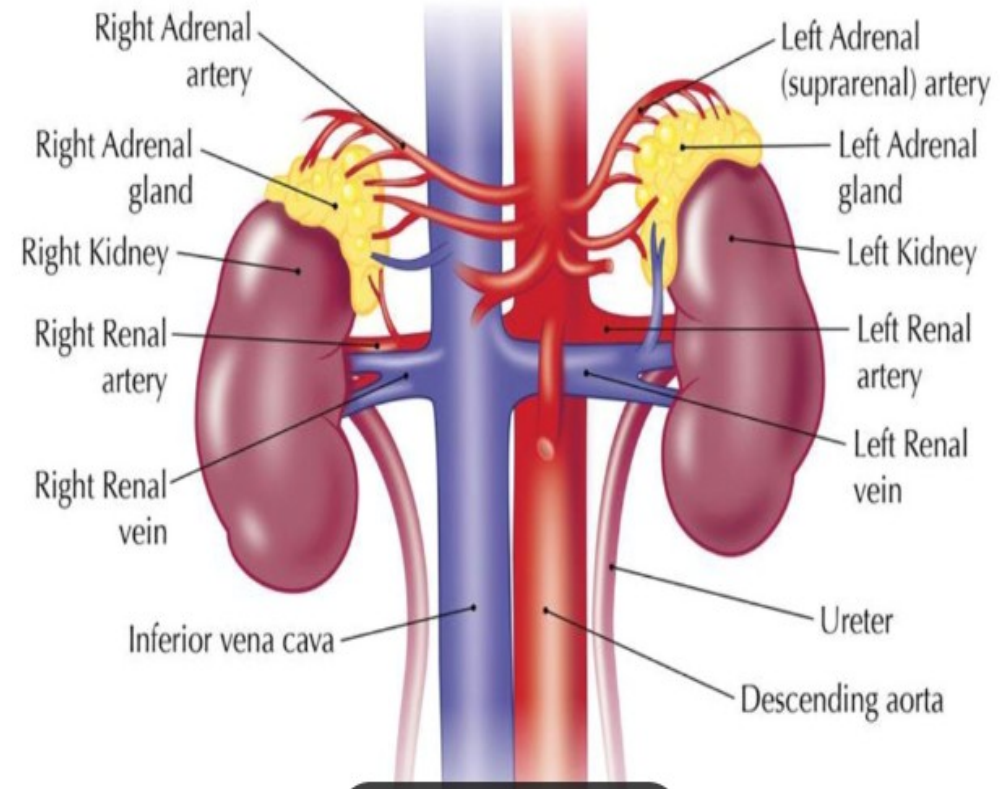
- **Central diabetes insipidus** is caused by **deficiency of ADH**, which results in
 - hypotonic polyuria, polydipsia,
 - hypernatremia, and dehydration.
 - Causes include head trauma and tumors.
- **Nephrogenic diabetes insipidus** is caused by **a lack of renal response to ADH**.

✓ **Syndrome of inappropriate ADH secretion (SIADH)**

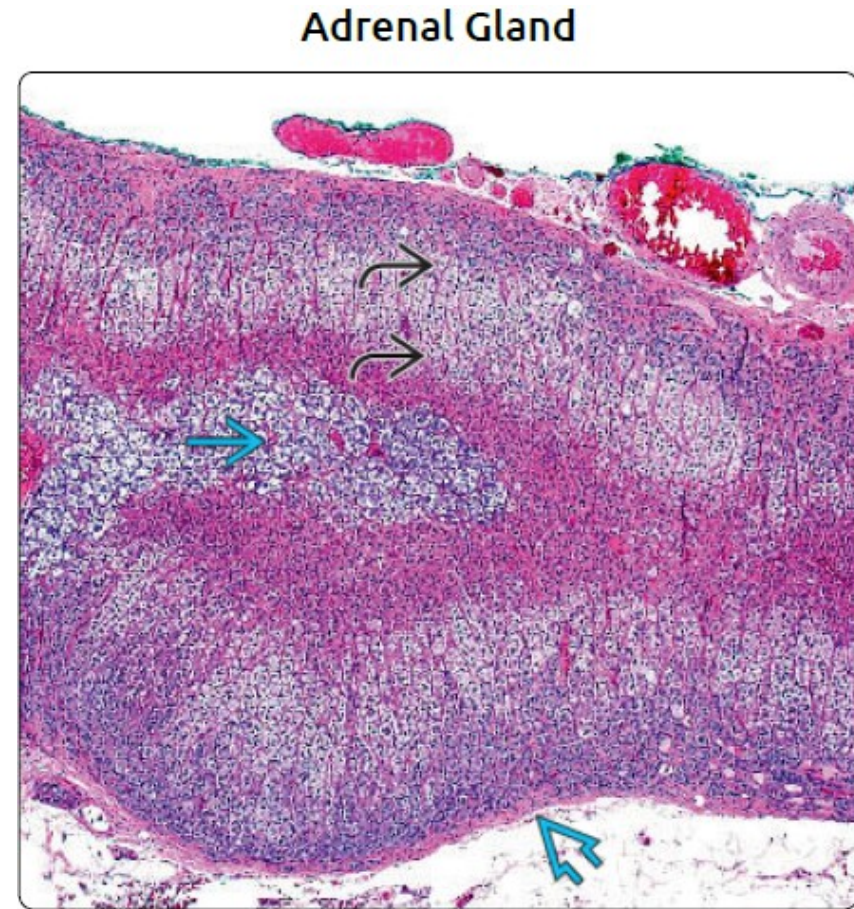
- Is caused by excessive production of antidiuretic hormone (ADH), resulting in oliguria, water retention, hyponatremia, and cerebral edema.
- Causes include **paraneoplastic syndrome** and **head trauma**.
- **Craniopharyngioma** is a benign pituitary tumor derived from **Rathke pouch** remnants that are usually located above the sella turcica, but can extend downward to destroy the pituitary.
- It is the most common cause of hypopituitarism in children.

ADRENAL GLAND

- The cells of the adrenal medulla are derived from **neural crest cells**.
- The cells of the adrenal cortex are derived from **mesoderm**



- Cortex
 - Glucocorticoids(cortisol)
 - Aldosterone
 - Steroid Hormones
- Medulla
 - Cathecolamine



Hyper adrenalism

✓ Cushing syndrome

- Is characterized by increased levels of **glucocorticoids**.
- The most common cause is **exogenous** glucocorticoid administration

✓ **Endogenous** causes include:

- **Cushing disease** (hypersecretion of ACTH, usually due to a pituitary microadenoma)
- Secretion of ACTH from nonpituitary tumors (e.g., small cell lung cancer)
- ACTH-independent Cushing syndrome due to adrenal neoplasia

Clinical manifestations of Cushing syndrome

- Include hypertension, weight gain (truncal obesity, “buffalo hump” and moon face's), cutaneous striae, hirsutism and mental disturbances

➤ **Hyperaldosteronism**

- May cause hypertension and hypokalemia.
- ✓ **Primary** (decreased plasma renin)
 - Adrenocortical neoplasm: adenoma (Conn syndrome) and Carcinoma
 - Bilateral nodular hyperplasia of the adrenal
- ✓ **Secondary** (increased plasma renin) (e.g., renal artery stenosis)

➤ **Adrenogenital syndromes**

- are adrenal disorders characterized by excess production of androgens and virilization.
- They are caused either by
 - Adrenocortical adenoma/carcinoma**: which produces androgens, or
 - Congenital adrenal hyperplasia**: cluster of autosomal recessive enzyme defects (most common is **21-hydroxylase deficiency**).

Adrenal Insufficiency

- ✓ **Waterhouse- Friderichsen syndrome** (acute adrenal insufficiency)
 - Is a potentially fatal, bilateral hemorrhagic infarction of the adrenal glands associated with sepsis, often due to an **N. meningitides** infection in children.
 - It is clinically characterized by disseminated intravascular coagulation (DIC), acute respiratory distress syndrome, hypotension and shock, and acute adrenal insufficiency.
 - Treatment is antibiotics and steroid replacement.

Addison's disease (chronic adrenocortical insufficiency)

- Is caused by destruction of the adrenal cortex, leading to a deficiency of glucocorticoids, mineralocorticoids, and androgens.
- The most common cause is [autoimmune adrenalitis](#), though adrenal involvement by TB or metastatic cancer are other possible causes.
- Patients present with gradual onset of weakness, skin hyperpigmentation, hypotension, hypoglycemia, poor response to stress, and loss of libido.
- Treatment is steroid replacement

N.B:- Secondary adrenocortical insufficiency may be caused by disorders of the hypothalamus or pituitary (cancer or infection, for example). Since ACTH levels are low, hyperpigmentation does not occur.

➤ **Pheochromocytoma** (“dark/dusky-colored tumor”)

- Is an uncommon benign tumor of the adrenal medulla, which produces **catecholamine's** (norepinephrine and epinephrine).
- It can present with sustained or episodic hypertension and associated severe headache, tachycardia, palpitations, diaphoresis, and anxiety

N.B:- Note the **rule of 10s** of Pheochromocytoma

- **10%** occur in children
- **10%** are bilateral
- **10%** are malignant
- **10%** are familial (MEN 2A and 2B)
- **10%** occur outside the adrenal gland
- Urinary **vanillylmandelic acid (VMA)** and **catecholamine** are elevated.
- Microscopically, the tumor shows nests of cells (**Zellballen**) with abundant cytoplasm.

ENDOCRINE DISORDERS IMPACTING MULTIPLE ORGANS

➤ Multiple Endocrine Neoplasia Syndromes

- Multiple endocrine neoplasia (MEN) syndromes are autosomal dominant conditions with incomplete penetrance
- Are characterized by hyperplasia and tumors of endocrine glands occurring at a young age.

➤ MEN 1 (Werner syndrome)

- Features tumors of the **p**ituitary gland, **p**arathyroids, and **p**ancreas
- Affected gene is **MEN1**, a tumor suppressor gene which encodes a nuclear protein called menin
- Associated with peptic ulcers and Zollinger -Ellison syndrome

3 P'S

➤ **MEN 2:**

- ✓ **MEN 2A** (Sipple syndrome) features **m**edullary carcinoma of the thyroid, pheochromocytoma (tumor of adrenal **m**edulla), and **p**arathyroid hyperplasia or adenoma.

MPM

- Mutation of **RET** proto-oncogene

- ✓ **MEN 2B** features **m**edullary carcinoma of the thyroid, pheochromocytoma (tumor of adrenal **m**edulla) , and **m**ucocutaneous neuromas.
- Mutation of **RET** proto-oncogene

3M's

Pancreas

Diabetes Mellitus

- Is a chronic systemic disease characterized by **insulin deficiency** or **peripheral resistance**, resulting in hyperglycemia and nonenzymatic glycosylation of proteins.
- Diagnosis is established by demonstrating either of the following:
 - Fasting glucose **>126 mg/dL** on at least 2 separate occasions
 - Positive glucose tolerance test
- A glycated hemoglobin (HbA1c) assay is used for certain patient populations. The diagnostic cutoff is **6.5%**.

Type 1 diabetes (T1D)

- represents 10% of cases of diabetes.
- Affects children and adolescents, usually age <20year
- Pathogenesis is a lack of insulin due to **autoimmune destruction of β cells (type IV hypersensitivity reaction)**
- Patients are absolutely dependent on insulin to prevent ketoacidosis and coma
- Thought to be caused by autoimmune reaction triggered by an infection (**Coxsackie B virus**) in a genetically susceptible individual

- T1D can present with polydipsia, polyuria, and polyphagia; dehydration and electrolyte imbalance; metabolic ketoacidosis; and coma and potentially death.
- Microscopically, lymphocytic inflammation involves the islets of Langerhans (insulitis), leading to loss of β cells and fibrosis of the islets.

Type 2 diabetes (T2D)

- ✓ represents 90% of cases.
- Affects obese individuals, both children and adults
- Risk factors include obesity, increasing age, and genetic predisposition
- Pathogenesis involves relatively reduced insulin secretion; peripheral insulin resistance is the term used for reduced tissue sensitivity to insulin due to decreased numbers of insulin receptors on the cell membranes.

- T2D is often asymptomatic, but it can present with either polydipsia, polyuria, and polyphagia, or with hyperosmolar non ketotic diabetic coma.
- Microscopically, the changes are nonspecific, and can include focal atrophy and amyloid deposition in islets (hyalinization).

complications of DM

➤ Acute

- Diabetic ketoacidosis
- Hypoglycemia
- Non ketotic hyperosmolar coma

➤ Chronic

✓ Microvascular

- Diabetic nephropathy
- Diabetic retinopathy
- Diabetic neuropathy

✓ **Macrovascular**

- **Cardiovascular disease** is the leading cause of death among diabetics.
- **Peripheral vascular disease** in diabetics is the leading cause of non traumatic amputations.

Disease of the CNS

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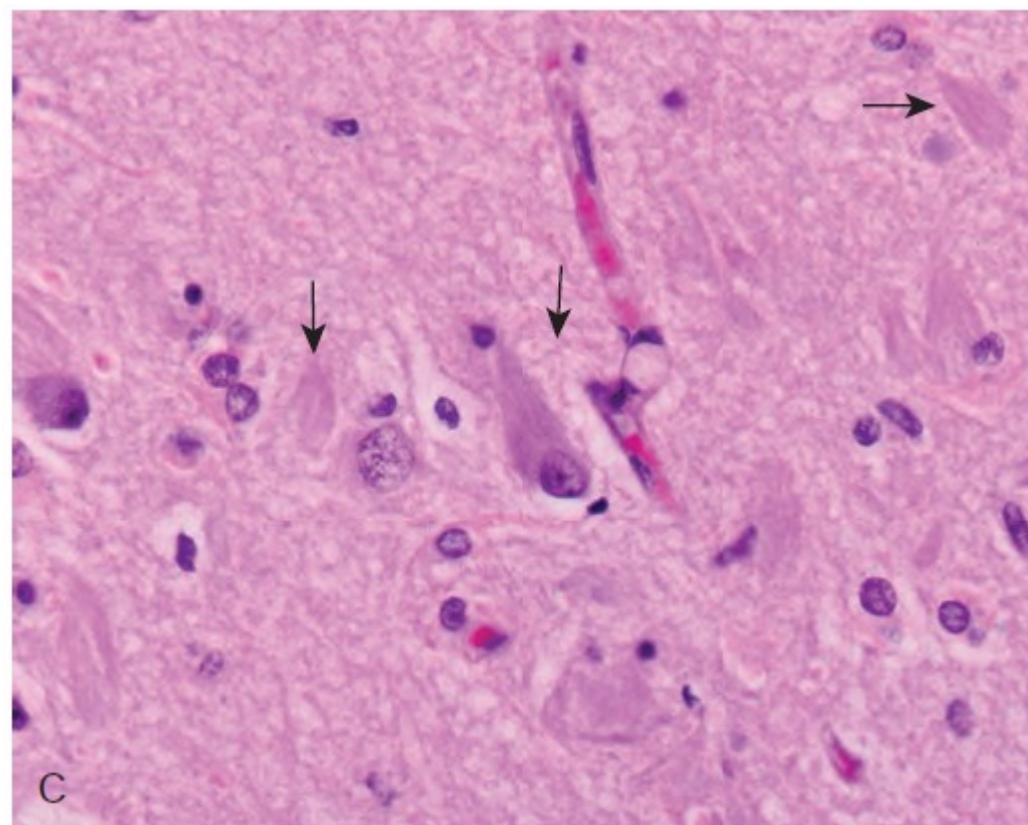
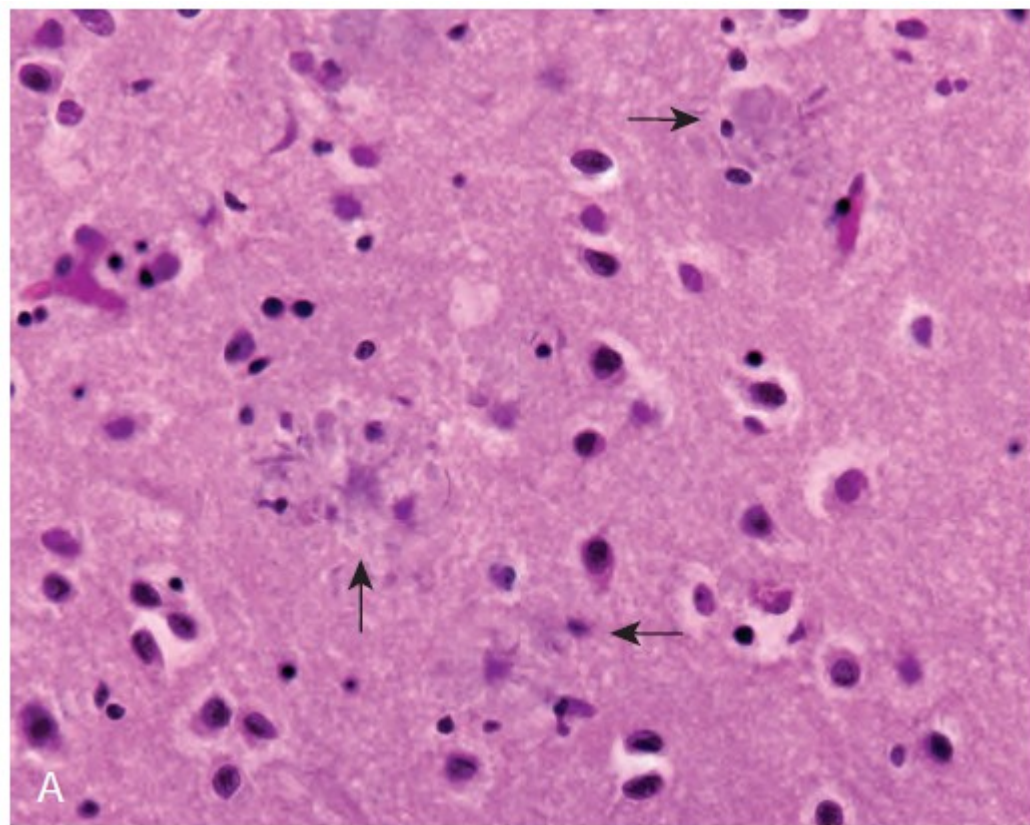
Neurodegenerative Diseases

Alzheimer Disease (AD)

- **most common cause of dementia in older adults, with an increasing incidence as a function of age.**
- The disease usually becomes clinically apparent as insidious impairment of higher cognitive functions.
- As the disease progresses, deficits in memory, visuospatial orientation, judgment, personality, and language gradually emerge;
- over a course of 5 to 10 years, the affected individual becomes profoundly disabled, mute, and immobile.
- **The note book**

Pathology

- **The fundamental abnormality in AD is the accumulation of two proteins (A β and tau) in specific brain regions, likely as a result of excessive production and defective removal.**
- The two pathologic hallmarks of AD, particularly evident in the end stages of the illness, are *amyloid plaques and neurofibrillary tangles*
- Plaques are deposits of aggregated A β peptides in the neuropil, while tangles are aggregates of the microtubule binding protein tau.
- The combination of clinical assessment and current radiologic methods allows accurate premortem diagnosis in 80% to 90% of cases.



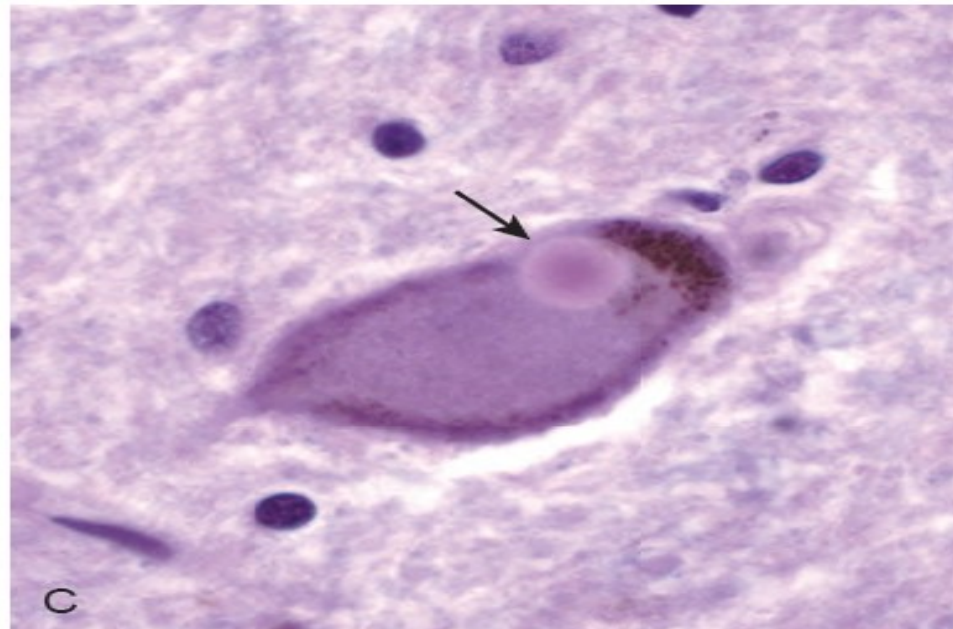
Parkinson Disease (PD)

- **neurodegenerative disease marked by a hypokinetic movement disorder that is caused by loss of dopaminergic neurons from the substantia nigra.**
- Nigrostriatal pathway of basal ganglia uses dopamine to initiate movement.
- The clinical syndrome of *parkinsonism* combines diminished facial expression (often termed masked facies), stooped posture, slowing of voluntary movement, festinating gait (progressively shortened, accelerated steps), rigidity, and tremor.
 - Tremor at rest and disappear with movement.
- similar symptoms can be induced by dopaminergic antagonists or by toxins that selectively damage the dopaminergic system!!!!!!!!!!!!!!
- **Mohammed Ali.**

- The clinical diagnosis of PD is based on the presence of the triad of parkinsonism
 - tremor,
 - rigidity, and
 - bradykinesia—in the absence of a toxic or other known underlying etiology

Pathogenesis

- loss of pigmented neurons in the substantia nigra
- **PD is associated with protein accumulation and aggregation, mitochondrial abnormalities, and neuronal loss in the substantia nigra and elsewhere in the brain.**
- This protein is a major component of the *Lewy body* (filaments are composed of α -synuclein), which is the diagnostic hallmark o



Dementia With Lewy Bodies

- About 10% to 15% of individuals with PD develop dementia, particularly with advancing age.
- most prominent histologic correlate is the presence of widespread Lewy bodies in neurons of the cortex and brainstem.

Amyotrophic Lateral Sclerosis (ALS)

- **progressive disorder in which there is loss of upper motor neurons in the cerebral cortex and lower motor neurons in the spinal cord and brainstem.**
- Loss of these neurons results in denervation of muscles, producing weakness that becomes profound as the disease progresses.
- Pathology
- **associated with degeneration of upper and lower motor neurons, often in association with toxic protein accumulation.**

Meningitis

Inflammation of the leptomeninges

Meninges consist of three layers (dura, arachnoid, and pia) that lie between the brain and the skull,

- Pia and arachnoid together are termed leptomeninges

Most commonly due to an infectious agent

- Group B streptococci, *E coli*, and *Listeria monocytogenes* (neonates)
- *N.meningitidis* (children and teenagers). *Streptococcus-pneumoniae* (adults and elderly), and *H influenza* (n on vaccinated infants)
- Coxsackievirus (children; fecal-oral transmission)
- Fungi (immunocompromised individuals)

- Presents with classic triad of headache, nuchal rigidity, and fever;
- photophobia, vomiting, and altered mental status may also be present,
- Diagnosis is made by lumbar puncture (sampling of CSF).
- Performed by placing a needle between L4 and L5 (level of the iliac crest).
- Spinal cord ends at L2, but subarachnoid space and cauda equina continue to S2.
- Layers crossed include skin, ligaments, epidural space, dura, and arachnoid.

- **CSF findings**

1. Bacterial meningitis—neutrophils with decrease in CSF glucose; gram stain and culture
2. Viral meningitis—lymphocytes with normal CSF glucose
3. Fungal meningitis—lymphocytes with decrease CSF glucose

Complications are usually seen with bacterial meningitis.

1. Death—herniation secondary to cerebral edema
2. Hydrocephalus, hearing loss, and seizures—sequelae related to fibrosis

CNS Tumors

- Can be metastatic (50%) or primary (50%)
- Metastatic tumors characteristically present as multiple, well-circumscribed lesions at the gray-white junction.
 - Lung, breast, and kidney are common sources.
- Primary tumors are classified according to cell type of origin (e.g., astrocytes, meningotheial cells, ependymal cells, oligodendrocytes, or neuroectoderm).
- In adults, primary tumors are usually supra tentorial.

Most common tumors in adults are glioblastoma multiforme, meningioma, and schwannoma.

- In children, primary tumors are usually infratentorial

Most common tumors in children are pilocytic astrocytoma, ependymoma, and medulloblastoma,

- Primary malignant CNS tumors are locally destructive, but rarely metastasize

Thank you!

Recap

Cell adaptation

- Hypertrophy
- Hyperplasia
- Atrophy
- metaplasia

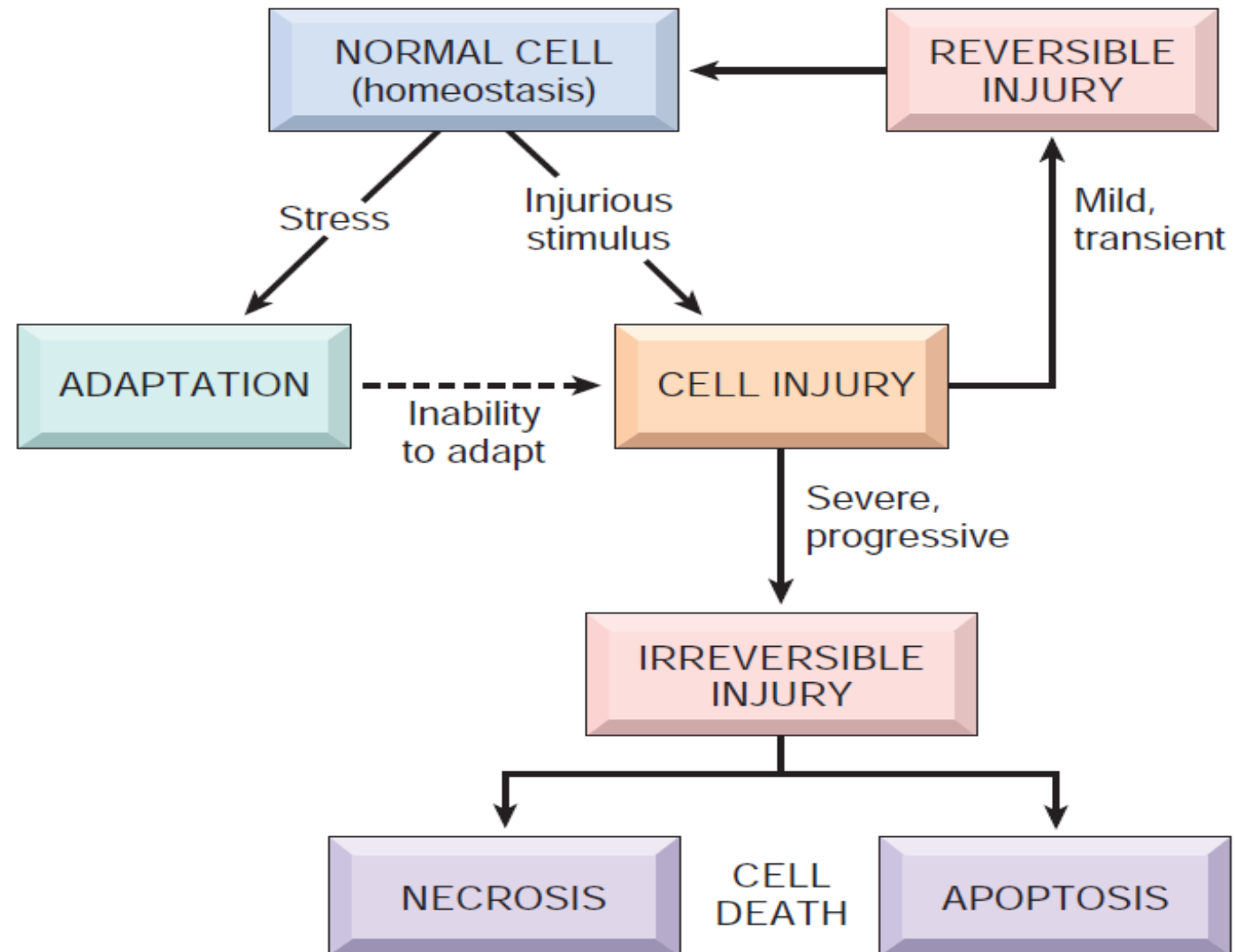


Figure 2.1 Stages of the cellular response to stress and injurious stimuli.

Reversible Cell injury

- **correctable if the damaging stimulus is removed**
 - *Generalized swelling of the cell and its organelles, blebbing of the plasma membrane, detachment of ribosomes from the endoplasmic reticulum (ER),*
 - *Fatty change*

- Irreversible injury is marked by :
 - severe mitochondrial vacuolization
 - extensive damage to plasma membranes
 - swelling of lysosomes
 - the appearance large, amorphous densities in mitochondria

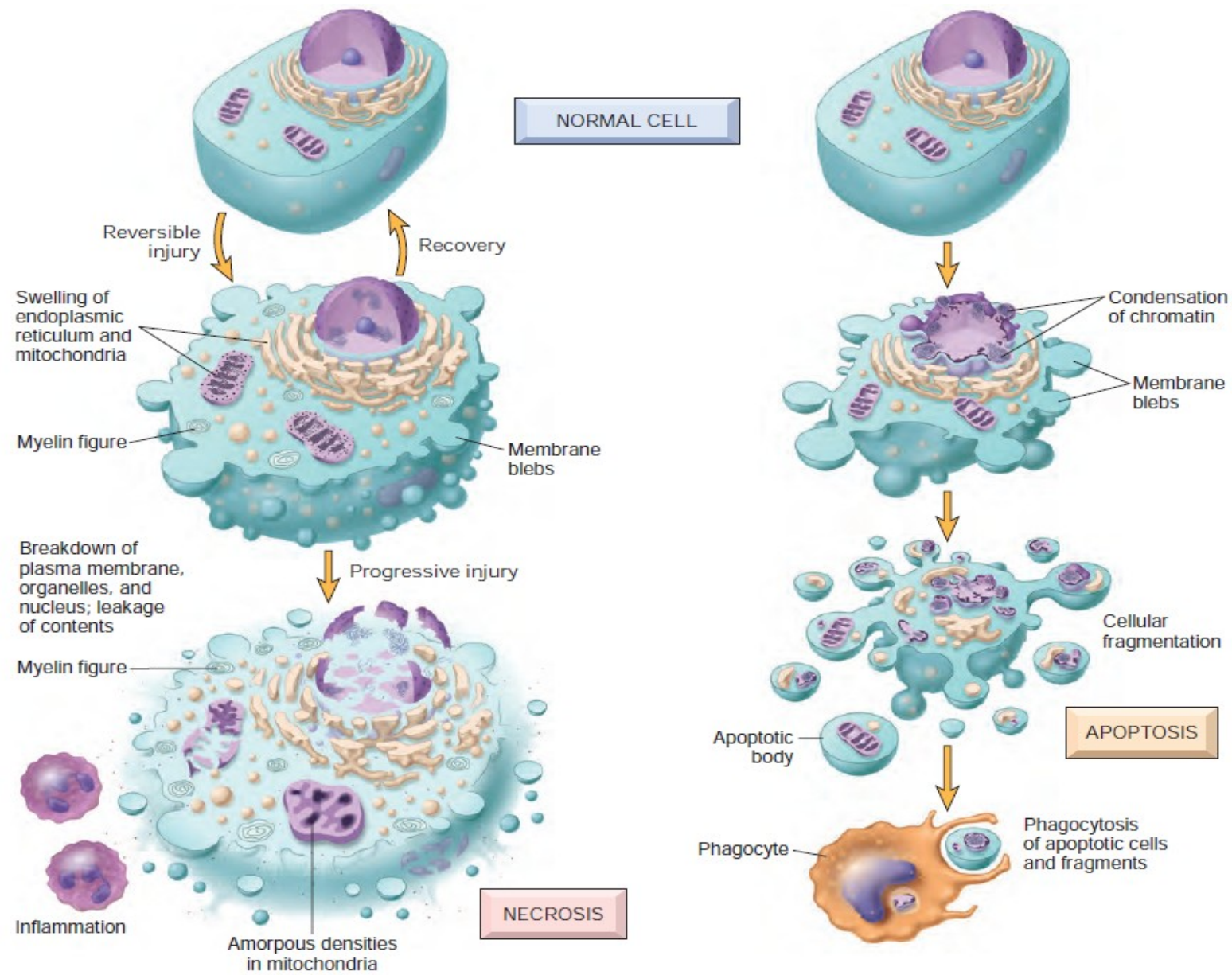


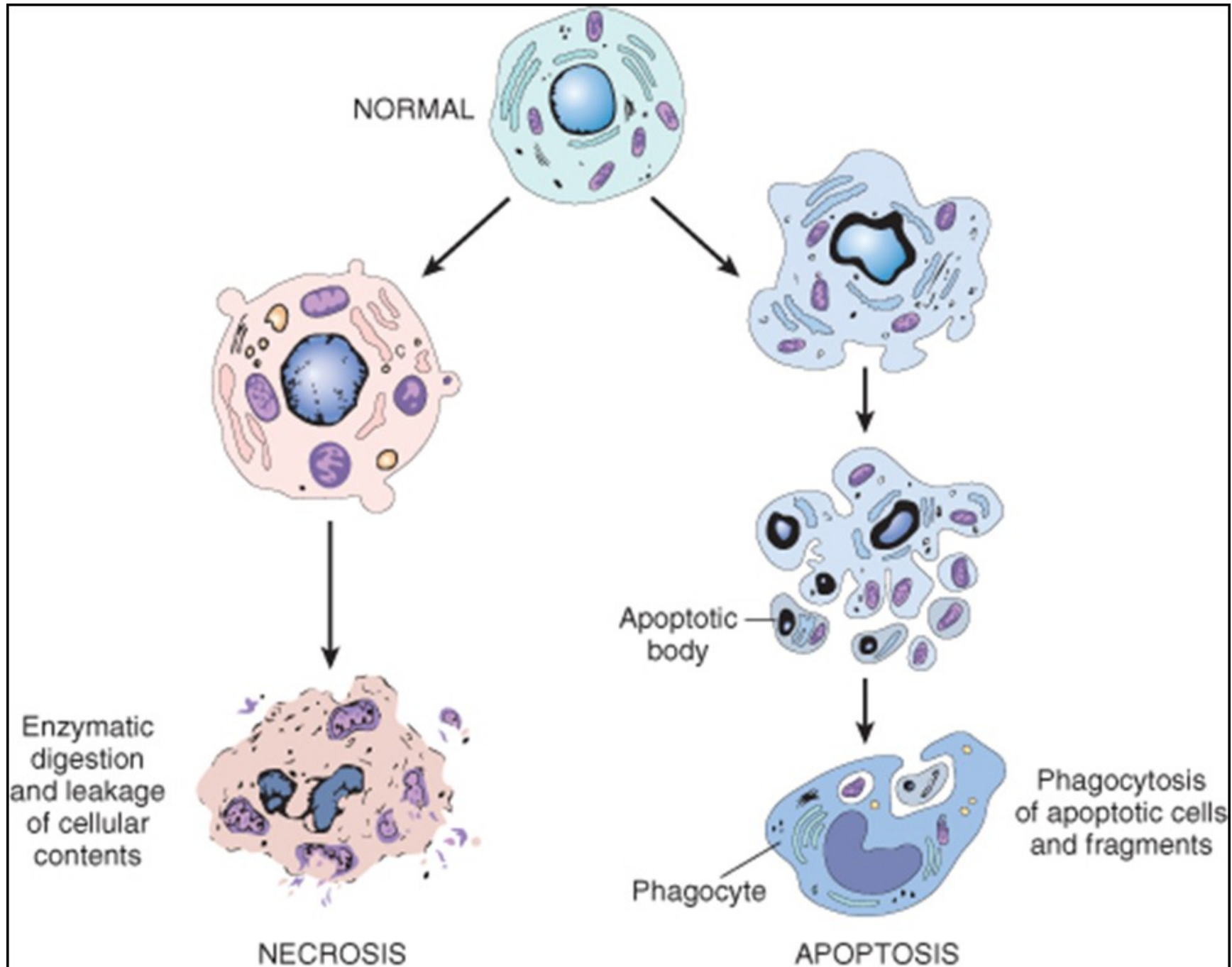
Figure 2.4 Schematic illustration of the morphologic changes in cell injury culminating in necrosis or apoptosis.

- Apoptosis

The cell's plasma membrane remains intact, but its structure is altered in such a way that the apoptotic cell sends signal to macrophages to phagocytose it.

- Necrosis

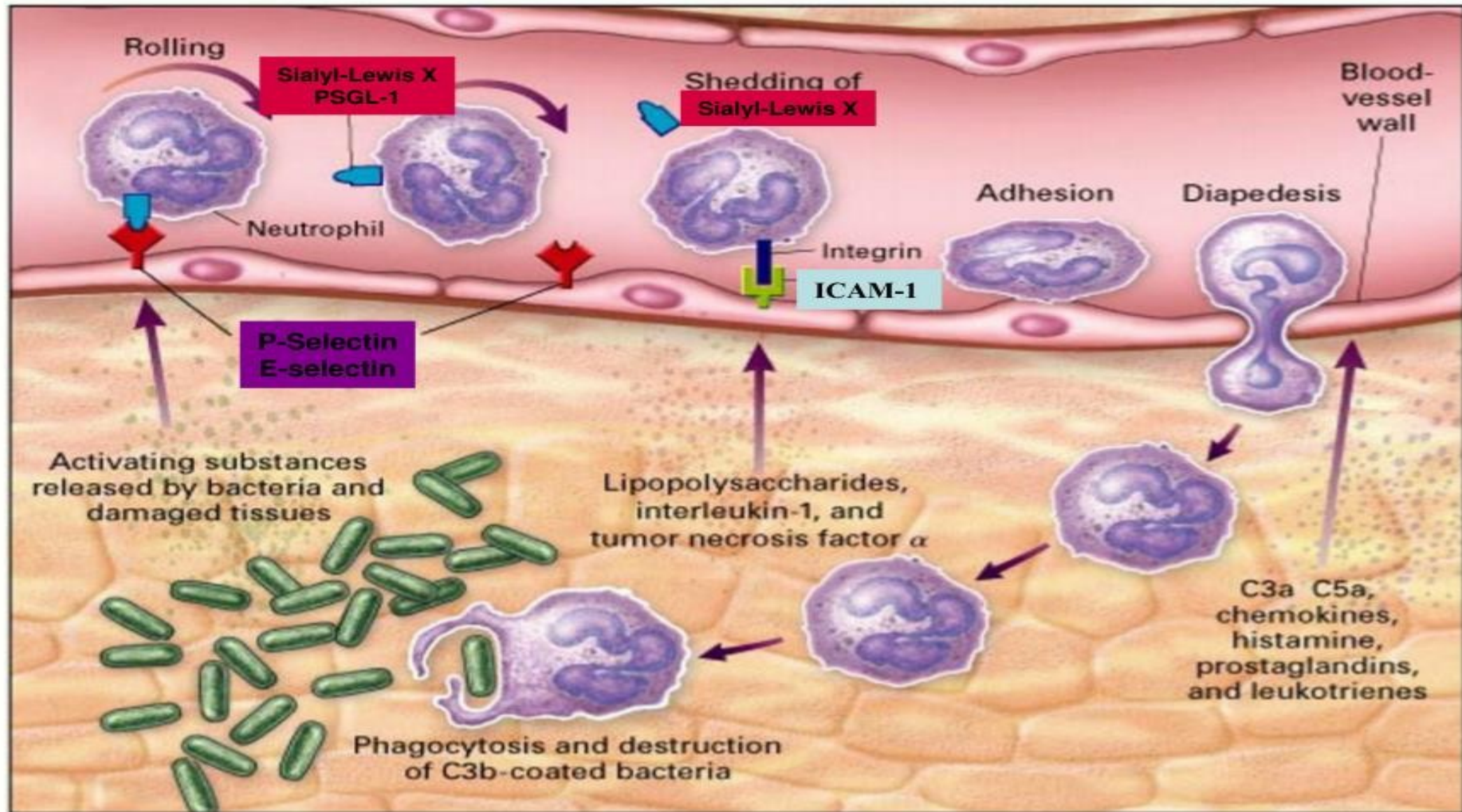
- Always pathological
- **leakage of cellular contents through damaged membranes, local inflammation, and enzymatic digestion of the lethally injured cell.**



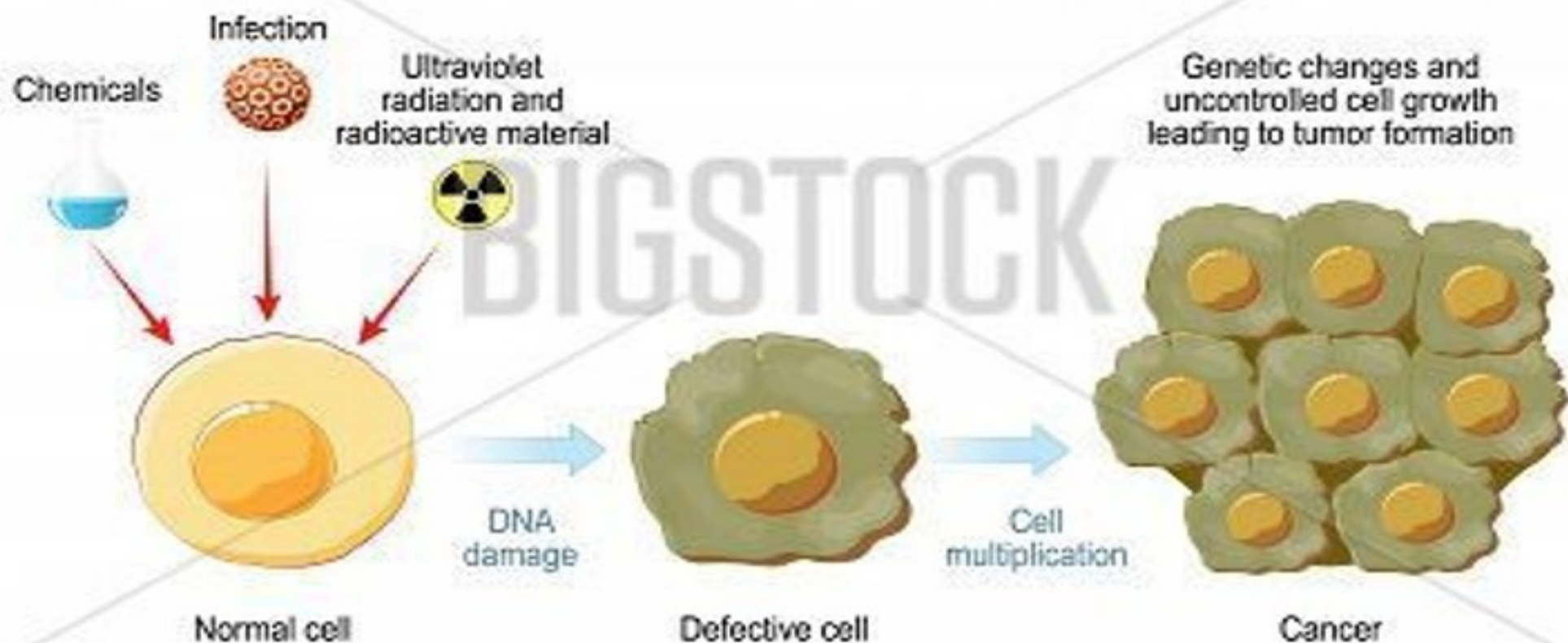
- Acute inflammation
immediate and early response
to an injurious agent
relatively of short duration,
lasting for minutes, several
hours or few days.
It is characterized by **exudation
of fluids** and plasma proteins
and the emigration of
predominantly **neutrophilic**
leucocytes to the site of injury

- Chronic inflammation
prolonged inflammatory
process (weeks or months)
➤ **Cells of chronic inflammation**
 - Monocytes and Macrophages
 - Lymphocytes

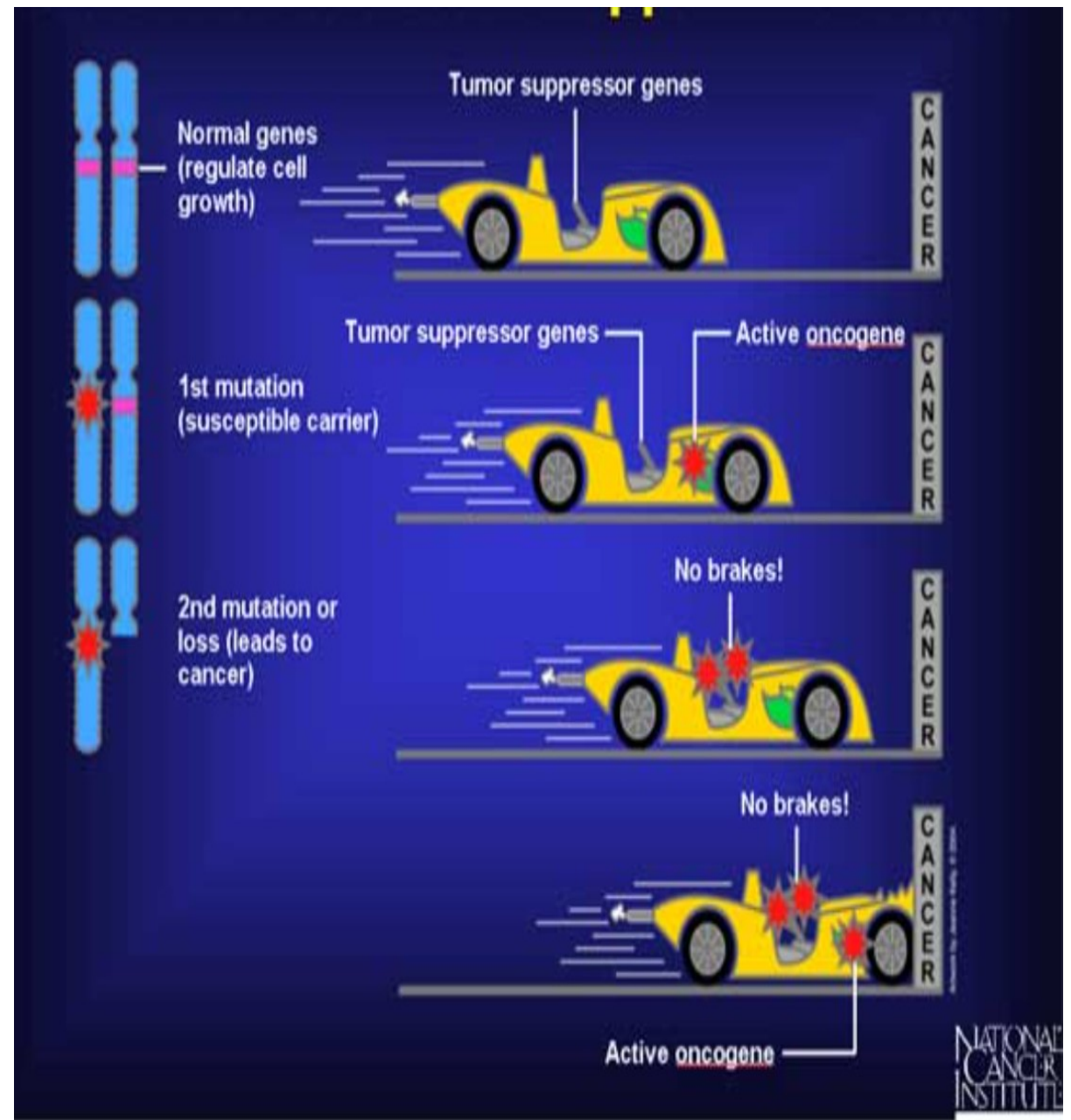
Neutrophil Recruitment: the Multistep Model



Process of cancer development



- The most important mutations in tumorigenesis involve:
 - growth promoting genes (**proto-oncogenes**),
 - growth inhibiting tumor **suppressor genes**,
 - the genes regulating **apoptosis** and **senescence**, and
 - **DNA repair genes**



- Benign vs malignant